MULTI-BRANCH RESTAURANT ORDERING & POS PLATFORM

Software Requirements Specification

Software Requirements Specification (SRS): Multi-Branch Restaurant Ordering & POS Platform

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Case study of a real project. Client details are withheld, and names and data are changed. Source and full traceability: github.com/ayushsingh0456/multi-branch-restaurant-ordering-pos-platform

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Purpose and scope. This SRS specifies what the platform does, how it interfaces with people and other systems, and the quality it must meet. It covers Release 1 (R1, live since 2026-06-29) and the increments R1.1 and R1.2. It is the contract between the Product Owner and the delivery team, and the basis for test design. Every requirement is written to be verifiable; section 4 says how.

1. Introduction

1.1 Purpose

This SRS turns the business needs BN-01 to BN-07 in the BRD into verifiable software requirements. Its audiences are:

- Developers and the Tech Lead, who design against it.
- QA, who derive test cases from it.
- The UX Designer, who builds screens to the field-level specifications.
- The Finance Controller and the Data Protection Officer, who check that rules with tax, fiscal or privacy weight are stated precisely.

The layout follows the SRS information item in ISO/IEC/IEEE 29148:2018. The overall description is kept as its own section (section 2) for readability.

1.2 Scope

The platform is the pickup-ordering and in-store till platform of the restaurant group, a chain of four grill-bowl and wrap restaurants in one EU member state. Four applications share one backend:

- **Customer Mobile App** (iOS and Android) and **Customer Web Ordering panel**: customers choose a branch, customize grill bowls and wraps, pick a pickup minute, pay online or at the counter, and follow the order live.

- **Counter Staff iPad till**: staff take walk-in and dine-in orders, settle by cash or card, and work the live queue of orders from every channel.
- **Admin Panel**: Administrators configure branches, menus and staff; Managers run the kitchen and counter boards of their branches.

This version specifies 72 functional requirements in 7 modules, 45 business rules (detailed in business-rules.md) and 28 non-functional requirements (detailed in non-functional-requirements.md).

Out of scope:

- Delivery to the customer's address.
- Loyalty, vouchers and marketing messages.
- Table service and self-order kiosks.
- Payroll, rota planning and stock control.

The reasons are in BRD section 5.3. The business outcomes the platform must move are OBJ-01 to OBJ-06 (BRD section 4).

1.3 Product overview

Application	Users	Technology	Release
Customer Mobile App	CUS, GST	Flutter (Dart), one code base for iOS and Android	R1
Customer Web Ordering panel	CUS, GST	React 18 with Redux, single-page app	R1
Counter Staff iPad till	CST, MGR	Flutter (Dart), iPad, landscape only	R1
Admin Panel (including the kitchen and counter boards)	ADM, MGR	React 18 with Redux	R1

All four front ends use one REST API (`/v1`, contract in openapi.yaml) served by a Node.js and Express backend with Mongoose on MongoDB. Live updates use Socket.IO (realtime-events.md).

1.4 Definitions, acronyms and conventions

Terms are defined in the glossary. Conventions used in this document:

Convention	Meaning
shall	A mandatory, verifiable requirement. Prose without "shall" is explanatory.
Priority	MoSCoW: Must, Should, Could, Won't (this release).
Release	R1 (pilot 2026-06-15 at LOC-01, all branches 2026-06-29), R1.1 (2026-09-07), R1.2 (2026-09-28).
IDs	FR--NN, BR-NNN, US-NNN, NFR--NN, IF-NN (interfaces, section 3.1.3), TBD-NN (open issues, Appendix A), WE-N (worked examples, Appendix B). IDs are never reused.
Time	24-hour clock in the branch's local time (CET/CEST). The server's clock is the only clock for slots and boards (BR-020). Stored in UTC; dates are ISO 8601.
Money	Euros in documents and the UI, with prices including VAT. The API carries integer cents (`amountCents`).
Kitchen minute	One minute of the branch's single production line. Pickup slots are calculated in kitchen minutes (BR-011, BR-012).
Defaults	Numbers in requirements are defaults. Section 2.7 lists them and who can change them.

1.5 References

Reference	Use in this SRS
BRD, business-rules.md, non-functional-requirements.md, compliance-mapping.md, glossary.md	Companion volumes of this specification
Requirements traceability matrix	BN to FR to BR to US to API to test case
Data dictionary, ERD	Entity and field definitions
openapi.yaml, API guidelines, realtime events	API contract, error codes, live events
State machines, sequence diagrams, wireframes	Behavior and screen design
ISO/IEC/IEEE 29148:2018	Requirements engineering; SRS structure; verification methods
ISO/IEC 25010:2011	Quality model for the NFRs
Regulation (EU) 2016/679 (GDPR)	Personal data, erasure, minimization
Council Directive 2006/112/EC (VAT Directive)	Invoice content; VAT per rate
Regulation (EU) No 1169/2011 (food information to consumers)	Allergen information before purchase (CR-001)
Directive (EU) 2015/2366 (PSD2) and its SCA rules	Strong customer authentication for online card payments
Directive (EU) 2019/882 (European Accessibility Act); WCAG 2.2; EN 301 549	Accessibility of the customer apps
PCI DSS v4.0; OWASP ASVS 4.0.3; RFC 9457	Card data scope, security verification, API error format

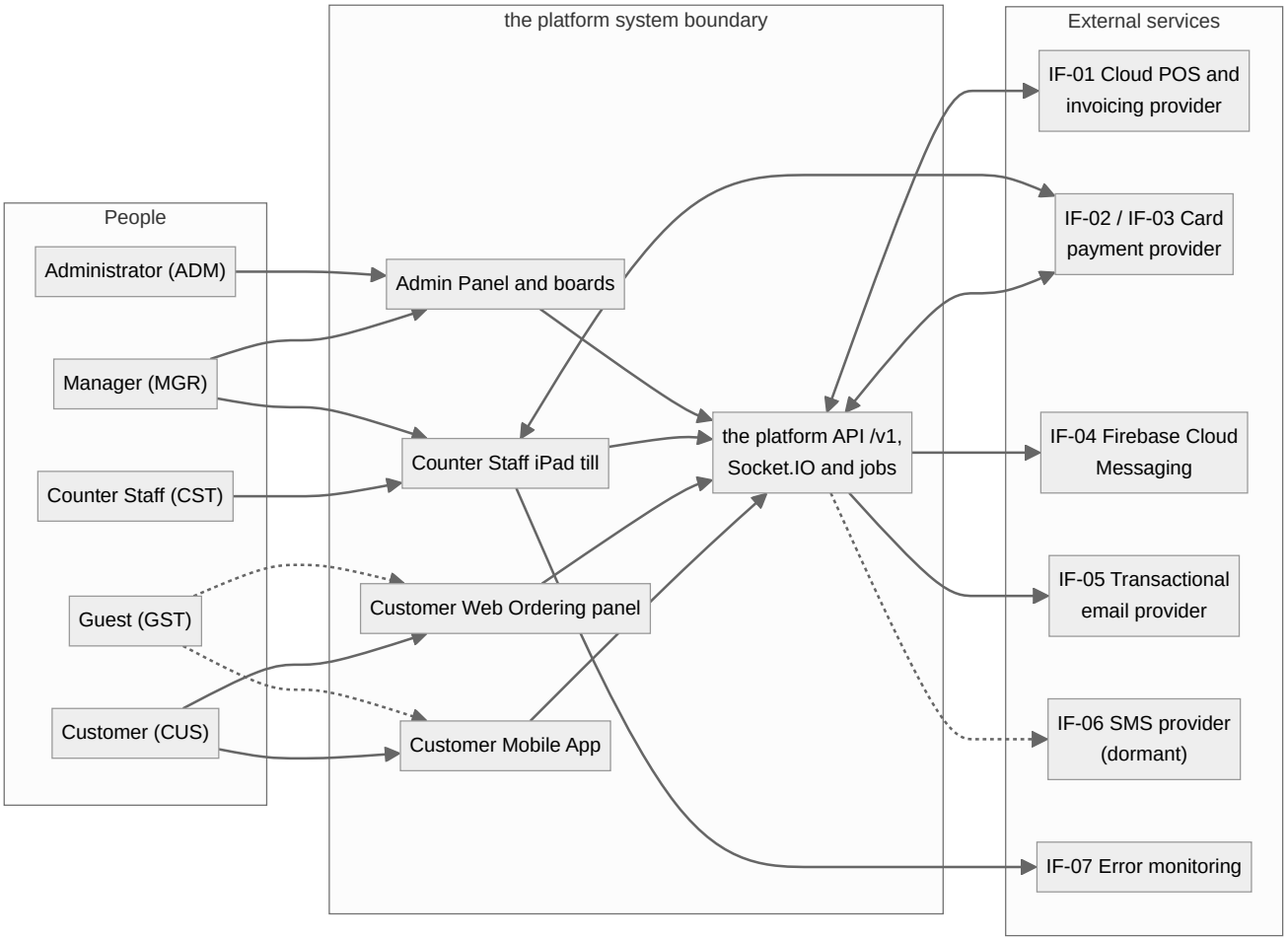
1.6 Revision history

Version	Date	Change
v1.0	2026-02-27	Baseline at the end of discovery.
v1.1	2026-04-10	CR-001: allergen information before purchase (FR-MNU-07, BR-028). CR-002 (guest checkout on the web panel) recorded as rejected; FR-IAM-03 unchanged.
v1.2	2026-06-05	UAT clarifications: one cancellation boundary for both customer apps ("at exactly 15:00 minutes before pickup is still allowed", BR-030); VAT extracted per rate group, not per line (BR-025, DEF-031); till drafts held per device (FR-TIL-04, DEF-038); paid orders cannot be called up (BR-034).
v1.3	2026-09-04	R1.1. CR-003: no-show leads to Prepay-only instead of suspension (BR-041, BR-042, FR-NTF-04). CR-004: resync after reconnect with branch sequence numbers (BR-045, FR-TIL-11; from INC-2026-004). CR-007: payment attempts with idempotency and asynchronous invoicing; nightly reconciliation (BR-035, BR-037, BR-039, FR-PAY-05, FR-PAY-06, FR-PAY-10; from INC-2026-009). CR-005 recorded as rejected.
v1.4	2026-09-18	R1.2. CR-006: rush-hour online capacity throttle (BR-018, FR-BRN-12), specified in spec 001. Appendix A refreshed.

2. Overall description

2.1 Product perspective

The platform is a new product. It replaces phone pre-orders written on paper, the standalone counter POS screen and the paper tickets passed to the kitchen, described in current-vs-future-state.md. The platform owns the order, the pickup slot and the live state of the kitchen. It does not own fiscal receipts or card data: invoices and receipts are raised in each branch's account at the cloud POS and invoicing provider, and card data stays with the card payment provider.



Container views, the tech stack and deployment are in system-architecture.md.

2.2 Product functions

Module	Epic	Functional requirements	Must	Should	Could
IAM	EP-01 Identity & Access	FR-IAM-01 to FR-IAM-10	9	1	0
BRN	EP-02 Branches & Pickup Slots	FR-BRN-01 to FR-BRN-12	11	1	0
MNU	EP-03 Menu & Catalog	FR-MNU-01 to FR-MNU-08	6	2	0
ORD	EP-04 Customer Ordering & Notifications	FR-ORD-01 to FR-ORD-12	11	0	1
NTF	EP-04 Customer Ordering & Notifications	FR-NTF-01 to FR-NTF-07	6	1	0

Module	Epic	Functional requirements	Must	Should	Could
PAY	EP-05 Payments & Invoicing	FR-PAY-01 to FR-PAY-10	9	1	0
TIL	EP-06 Till & Live Order Boards	FR-TIL-01 to FR-TIL-13	11	2	0
Total		**72**	**63**	**8**	**1**

The chain runs through the modules:

- MNU (what can be sold) and BRN (when the kitchen can make it) feed ORD and TIL.
- ORD and TIL create orders; PAY settles and invoices them.
- TIL's boards move orders to Ready; NTF tells the customer and chases uncollected orders.
- IAM and the live-update layer (FR-TIL-10, FR-TIL-11) cut across all modules.

2.3 User classes and characteristics

Code	Role	Description	Primary interface	Data scope
ADM	Administrator	Head-office staff (Operations Director, Finance Controller) who configure branches, menus, staff and payments.	Admin Panel	All branches; customer accounts; reconciliation
MGR	Manager	Shift and branch managers who run the boards, set daily busy windows and sold-out items, and can also use the till.	Admin Panel boards; till	Assigned branches
CST	Counter Staff	Employees who take walk-in and dine-in orders and work the live queue on the till.	Till	Assigned branches; no Admin Panel access
CUS	Customer	A person with a verified email who orders for pickup.	Mobile app; web panel	Own account and orders
GST	Guest	A visitor who browses without signing in.	Mobile app; web panel	Public menu data
SYS	System	Scheduled and event-driven jobs: slot engine, reminders, no-show cutoff, outbox, reconciliation.	Background jobs	Per job, branch-scoped where relevant

What the user classes are like, and what that means for design:

- **Customers (PER-01 Clara Mendes, PER-02 Tomasz Nowak)** order in a short lunch break, often on mobile data, and decide in seconds whether a pickup time works. They need an honest pickup time, a visible cancellation deadline and allergen information before they commit (NFR-USE-01).
- **Counter Staff (PER-03 Amir Haddad)** work a queue of walk-ins while online orders arrive. Every extra tap at the counter costs throughput at the busiest hour (NFR-USE-02). They are not trusted with refunds.
- **Kitchen leads (PER-04 Sofia Rinaldi)** read the kitchen board at arm's length, with gloves on, in a hot and noisy room. The board must be legible without color and must never silently go stale (NFR-ACC-02, NFR-REL-02).
- **Managers (PER-05 Marta Kowalska)** react to what happens on the floor: a fryer failure, a sold-out item, a late order. They need fast, branch-scoped controls.
- **Administrators (PER-06 Paul Lindqvist)** configure four branches and own money and compliance questions. They need audit trails and reconciliation, not live screens.

2.4 Operating environment

Component	Environment
Customer Mobile App	iOS 16+ and Android 10+; usable on mobile data at 1 Mbps; forced update below the minimum supported version (NFR-MNT-02)
Customer Web Ordering panel, Admin Panel	Last two major versions of Chrome, Edge, Firefox and Safari; boards run full-screen on a 24-inch display in the kitchen and at the counter
Counter Staff iPad till	iPadOS 16+ in landscape; the card payment provider's card reader paired over Bluetooth; branch Wi-Fi with 4G hotspot fallback
Backend	Node.js LTS with Express and Mongoose; MongoDB replica set (multi-document transactions); Socket.IO with a shared adapter across API instances; managed cloud hosting in an EU region (NFR-PRIV-03)
Time zone	All four branches use the same local time zone (CET/CEST); the server converts at the edge

2.5 Design and implementation constraints

Constraint	Source
The server is authoritative for prices, VAT and pickup slots. Clients never send prices or decide availability (BR-027, BR-012).	ADR-001
Each kitchen minute of a branch can be held by one order only; this is enforced by a unique index on the reservation, not only by a check in code.	ADR-001
Socket.IO events tell clients that something changed; the versioned branch snapshot over REST is the source of truth after any reconnect (BR-045).	ADR-002
Payments are attempts with an idempotency key; invoices are raised through an outbox, never inside the payment request (BR-035, BR-037).	ADR-003
REST under `/v1`; JSON with camelCase; cursor pagination; RFC 9457 problem details; `Idempotency-Key` on unsafe POSTs.	API guidelines
Card data never enters the platform: online payments use the provider's hosted checkout and counter payments use the provider's card reader SDK (NFR-SEC-03).	Compliance mapping
Fiscal receipts and invoices are raised only in the branch's POS account; the platform stores the reference and PDF link.	IF-01

2.6 Assumptions and dependencies

These are tracked with owners in the RAID log. The ones that would change the meaning of a requirement if they turned out false:

- Each branch has one production line. A kitchen minute is the unit of capacity (TBD-08 covers a second line).
- One legal entity owns all four branches, so one online merchant account at the card payment provider serves all branches (TBD-07).
- The cloud POS provider meets the national cash-register and receipt rules for every invoice it raises. The platform sends complete lines and VAT; it does not sign receipts itself.
- Tips are not part of the taxable order total; the Finance Controller confirmed this with the external tax adviser on 2026-05-14 (A-04).
- Branch Wi-Fi with a 4G fallback keeps the till online. The till has no offline mode in R1 (TBD-10).

2.7 Configuration defaults

Who can change a value: ADM = Administrator for any branch; MGR = Manager for assigned branches; Fixed = platform constant changed only by release.

Setting	Default	Range	Changed by
Customer one-time code	6 digits, valid 10 min, single use	Fixed	Fixed
Code resend delay; codes per email	60 s; 5 per rolling hour	Fixed	Fixed
Wrong entries per code	5, then the code is invalid	Fixed	Fixed
Staff password	12 characters minimum, breached-password check	Fixed	Fixed
Staff lockout	5 consecutive failures, 15 min	Fixed	Fixed
Staff password reset link	30 min, single use	Fixed	Fixed
Administrator second factor	TOTP, mandatory	Fixed	Fixed
Admin Panel idle sign-out	30 min; boards in board mode exempt, 14 h maximum session	Fixed	Fixed
Till session	14 h maximum, one per device	Fixed	Fixed
Weekly hours (LOC-01 to LOC-03)	Mon-Fri 11:00-21:00, rush 11:45-13:30; Sat 12:00-21:00, rush 18:00-19:30; Sun closed	Any chronological values (BR-009)	ADM, MGR
Weekly hours (LOC-04 Campus)	Mon-Fri 11:00-17:00, rush 11:45-13:30; Sat-Sun closed	As above	ADM, MGR
Online lead time L	10 min	5-60 min	ADM, MGR
Cancellation cutoff C	15 min before pickup	0-60 min	ADM, MGR
Minutes per kitchen unit m	1.0	0.5-5.0 in steps of 0.5	ADM
Rush factor f	0.5	0.25-1.0	ADM
Online order limit	30 kitchen units	10-60	ADM
Line quantity	1-20 in every channel	Fixed	Fixed
Busy window	One per branch per day, today only	Within opening hours	ADM, MGR
Delay protection cap	20 min	0-30 min	ADM
Rush throttle (R1.2)	Online share 60% of each 15-min rush block; reserve released 20 min before the block	Share 40-100%; release 10-45 min	ADM
Prepay-only slot hold	8 min	Fixed	Fixed
Online tip options	0%, 5%, 10%, 15%; 0% preselected	Fixed	Fixed
Counter card tip limit	Entered amount at most 1.5 x the order total	Fixed	Fixed
Reminder ladder	Push at anchor + 10 min; email at + 15 min; final email at + 30 min	Fixed	Fixed
No-show cutoff	Closing time + 30 min	Fixed	Fixed
Prepay-only duration	30 days from the no-show	Fixed	Fixed
Notification retries	3, at 1, 5 and 15 min after the first attempt	Fixed	Fixed
Live connection heartbeat; stale indicator	10 s; Reconnecting shown after 10 s without heartbeat	Fixed	Fixed

Setting	Default	Range	Changed by
Order history shown to customers	90 days, older by date filter (12 months)	Fixed	Fixed
Nightly reconciliation	03:00 local time, results before 08:00	Fixed	Fixed
Financial record retention	7 years (synthetic default; statutory period confirmed by the Finance Controller)	Fixed	Fixed
Synthetic VAT rates	Food 10% (takeaway and dine-in); drinks 20%	Per product	ADM

3. Specific requirements

3.1 External interface requirements

3.1.1 User interfaces

Screen designs are in the wireframes. These conventions apply to every screen:

Convention	Requirement
Status display	Order and payment states are shown with a text label and an icon, never color alone (WCAG 2.2 SC 1.4.1). Board tints for dine-in and employee color are supplementary and always carry a label or initials (NFR-ACC-02).
Validation messages	Inline at the field and summarized on submit. Each message states the problem and the fix (NFR-USE-03). The API returns RFC 9457 problem details with a stable code; codes are listed in api-guidelines.md and mapped to messages in section 3.2.
Prices	Shown in euros including VAT, with the locale's decimal separator. Net and VAT appear in the cart, order detail and invoice.
Times	Pickup times are whole minutes in 24-hour format. A deadline is shown as a clock time ("Cancel until 12:16"), not a countdown alone.
Destructive actions	Confirmation states the consequence, for example "Cancel order SQ-0312? EUR 22.68 will be refunded to the card."
Language	National language (default) and English in the customer apps (NFR-L10N-01). Staff apps use the national language with English available.
Mobile ergonomics	Touch targets at least 44 × 44 pt; one primary action per screen; supports 200% text scaling and screen readers (NFR-ACC-01).
Till ergonomics	Product tiles at least 96 × 72 pt; the cash button needs a 0.6 s press and hold so a stray tap cannot settle an order.

3.1.2 Hardware interfaces

Device	Used for	Requirement	When unavailable
iPad (landscape)	Till	Runs the till in landscape only; the screen does not sleep while the till is open.	Staff use another iPad; drafts are per device (FR-TIL-04), so nothing transfers.
Card reader (card payment provider)	Counter card payments (FR-PAY-04)	Paired over Bluetooth through the provider's SDK; the merchant signs in on the device once per day.	Cash or "pay later" (unpaid) remain available.

Device	Used for	Requirement	When unavailable
Kitchen and counter displays	Boards (FR-TIL-07, FR-TIL-08)	24-inch display, browser in full-screen board mode, sound enabled after a first interaction.	Any device with a browser can open the board; the live indicator shows the state (FR-TIL-11).
Receipt printer	Fiscal receipts	Owned by the POS provider's integration, outside the platform.	Customer receives the invoice in the app (FR-PAY-07).
Customer phone	Push, camera for profile picture	Push only after the customer allows notifications (FR-NTF-01).	Email reminders still run (BR-040).

3.1.3 Software interfaces

ID	Interface	Purpose	Direction	Data exchanged	Protocol	Failure handling
IF-01	Cloud POS and invoicing provider (one account per branch)	Payment methods import (FR-BRN-05); article sync (FR-MNU-06); invoices and cancellation receipts (FR-PAY-06, FR-PAY-09)	Both: API calls out; OAuth authorization per branch	Articles (name, price, VAT rate); invoice lines, VAT per rate, tip line, payment method; invoice number and PDF link. No customer contact data.	HTTPS REST with OAuth 2.0 tokens stored encrypted (NFR-SEC-04)	Invoices go through an outbox and retry with backoff until confirmed; idempotent by order ID (BR-037). An outage never blocks ordering or payment (NFR-REL-04). Expired tokens raise an Administrator alert.
IF-02	Card payment provider: online	Hosted checkout with SCA (FR-PAY-01); payment webhooks (FR-PAY-03); refunds (FR-PAY-08); transaction report (FR-PAY-10)	Both: API calls out; signed webhooks in at `POST /v1/webhooks/payments`	Amount, currency, attempt ID as merchant reference, return URL. Back: status, provider transaction ID, card brand and last 4 digits.	HTTPS REST with idempotency keys; webhook signature and timestamp checked	Webhooks processed once per provider event ID. A checkout with no webhook after 15 min is resolved by a status lookup. Refunds retry 3 times, then go to the Manager queue.
IF-03	Card payment provider: card reader SDK	Card-present payments on the till (FR-PAY-04)	Till to reader to provider	Amount and attempt ID; result and provider transaction ID	Provider SDK over Bluetooth; result verified server-side against the provider API (BR-038)	A timeout never triggers a new charge; the till offers "Verify payment" (BR-035, CR-007).

ID	Interface	Purpose	Direction	Data exchanged	Protocol	Failure handling
IF-04	Firebase Cloud Messaging	Push to the mobile app and web push (FR-NTF-01, FR-NTF-02, FR-NTF-03)	Out	Device token; title, short body with order number, branch and pickup time; deep link. No item details.	HTTPS (FCM HTTP v1)	Rejected tokens are deleted (BR-043). Email reminders do not depend on push.
IF-05	Transactional email provider (SMTP)	Sign-in codes, order confirmations, reminders, no-show notices, staff password reset (FR-NTF-05)	Out; bounce events in	Recipient, template, order number, branch, pickup time, payment link	SMTP over TLS; SPF, DKIM and DMARC aligned	Retries per BR-044. A hard bounce flags the address on the customer record.
IF-06	SMS provider (dormant)	Phone-number sign-in, built but not exposed	Out	Phone number and code	HTTPS REST	Disabled by feature flag in R1; no requirement depends on it. Kept to avoid rework if phone sign-in is reopened (DEC-09).
IF-07	Error monitoring service	Crash and error reports from the iPad till (NFR-OBS-01)	Out	Stack traces, release tag, device model, branch code. No customer data.	HTTPS SDK	Best effort; never blocks the till. Customer apps rely on the app stores' crash reports (TBD-04).

3.1.4 Communications interfaces

- All traffic uses HTTPS with TLS 1.2 or later and HSTS (NFR-SEC-01).
- Live updates use Socket.IO over WebSocket with long-polling fallback. Clients join rooms after authentication: `branch:{branchId}` for staff, `customer:{customerId}` for customers. The server checks the role matrix on every join (FR-IAM-06).
- Every branch event carries a monotonically increasing branch sequence number. Clients load `GET /v1/branches/{branchId}/snapshot` after every connect, reconnect or gap (BR-045).
- Unsafe POSTs carry an `Idempotency-Key`. A replay returns the original response.
- Push and email carry no item details or payment data (NFR-PRIV-01).

3.2 Functional requirements

Each module subsection gives a description, the epic, the actors, the requirement table and, where relevant, notes on edge cases and field-level specifications. In the requirement tables:

- **Related business rules** lists every BR that names the FR in business-rules.md.
- **Roles** lists the user classes that act in, or are served by, the requirement.

Field-level specifications define what the UI validates. The API enforces the same rules and returns RFC 9457 problem details with the error codes in api-guidelines.md. Messages state the problem and the fix (NFR-USE-03); example values are illustrative.

3.2.1 IAM: Identity & Access

Customer sign-in with an emailed one-time code, staff sign-in, roles, employee and customer administration, and account self-service.

Epic: EP-01 Identity & Access. **Actors:** Customer (CUS), Guest (GST), Counter Staff (CST), Manager (MGR), Administrator (ADM), System (SYS). **User stories:** US-001 to US-007.

ID	Requirement	Priority	Release	Roles	Related business rules
FR-IAM-01	The system shall let a customer sign in to the Customer Mobile App and the Customer Web Ordering panel with an email address and a one-time code sent to that address, and shall create the customer account on the first successful verification.	Must	R1	CUS	BR-001, BR-002
FR-IAM-02	The system shall enforce the one-time code limits: 6 digits, 10-minute validity, single use, invalidation by a newer code, a 60-second resend delay, 5 codes per email per rolling hour and 5 wrong entries per code.	Must	R1	CUS, SYS	BR-002
FR-IAM-03	The system shall let a guest browse branches, menus, prices and allergens in both customer apps without signing in, and shall ask the guest to sign in before adding an item to the cart.	Should	R1	GST	—
FR-IAM-04	The system shall authenticate Admin Panel users by email and password, require a TOTP second factor for Administrators, and refuse customer accounts.	Must	R1	ADM, MGR	BR-001, BR-006
FR-IAM-05	The system shall let Counter Staff and Managers sign in to the till with their own account, select one of their assigned active branches, and keep one session per device for up to 14 hours.	Must	R1	CST, MGR	BR-004
FR-IAM-06	The system shall enforce the role-permission matrix in section 3.3 server-side on every request and on every live-update subscription, including branch scoping for Managers and Counter Staff.	Must	R1	SYS	BR-003
FR-IAM-07	The Administrator shall be able to create, view, edit and deactivate employee accounts with name, email, role (fixed after creation), assigned branches and board color, and reset an employee's password without knowing it.	Must	R1	ADM	BR-001, BR-003, BR-005, BR-007
FR-IAM-08	The Administrator shall be able to search and view customer accounts, suspend and reinstate them, and clear a Prepay-only status, with every change recorded in the audit log.	Must	R1	ADM	BR-041
FR-IAM-09	A signed-in customer shall be able to edit their name and mobile number, sign out of the current device, and delete their account.	Must	R1	CUS	BR-008, BR-043
FR-IAM-10	The system shall let a staff member reset a forgotten Admin Panel password through a single-use emailed link valid for 30 minutes, without revealing whether the email is registered.	Must	R1	ADM, MGR	BR-006

Notes and edge cases:

- **No passwords for customers.** The account is created at the first successful code entry. A new account has no name or phone; checkout asks for a first name so the counter can call the order (field spec below).

- **Enumeration.** The code request returns the same response whether or not the email exists, except for a suspended account, which is told to contact the branch. A staff email entered in a customer app gets the same neutral response and no code (BR-001). Discovery found that the old flow told callers "this email is not available", which exposed staff addresses.
- **Code lifetime.** The previous system displayed "valid for 10 minutes" but never enforced it. The platform enforces it server-side (TC-IAM-002).
- **Role is fixed.** Changing an employee from Counter Staff to Manager means deactivating the account and creating a new one, so audit history never mixes privilege levels (BR-003, DEC-04).
- **Deactivation is immediate.** The old system let deactivated employees keep signing in. The platform revokes refresh tokens and disconnects live sessions within 60 seconds (BR-005).
- **Sign-out on one device.** Removing the push token of the current device only (BR-043). The previous system cleared push for every device of the account.
- **Kitchen board session.** In R1 the kitchen display runs under the shift Manager's board-mode session. A board-only role is TBD-01.

Field specification: customer sign-in (both apps)

Field	Type	Required	Validation	Message shown to user
Email address	Email	Yes	Valid syntax; trimmed; stored lowercase; at most 254 characters	"Enter an email address in the form name@example.com."
Send code	Action	n/a	At most 5 codes per email per rolling hour (BR-002)	"You have asked for 5 codes in the last hour. Try again at 12:41 or use a different email."
One-time code	6 numeric boxes; paste fills all	Yes	Exactly 6 digits; not expired; not used; fewer than 5 wrong entries	"Enter all 6 digits from the email." / "This code is not correct. You have 3 tries left." / "This code has expired. Send a new code." / "Too many wrong codes. Send a new code."
Resend code	Action with countdown	n/a	Enabled 60 s after the last send; sending invalidates the previous code	Button reads "Send a new code in 0:42" until enabled.
Suspended account	n/a	n/a	Account status Suspended	"This account is suspended. Contact the branch or support@example.com."
First name (asked at first checkout)	Text, 1-40	Yes, before the first order	Letters, spaces, apostrophes and hyphens	"Enter the first name we should call at the counter."

Field specification: employee account (Admin Panel)

Field	Type	Required	Validation	Message shown to user
Name	Text, 2-60	Yes	Letters, spaces, apostrophes, hyphens	"Enter the employee's full name."
Email	Email	Yes	Valid syntax; unique across staff and customer accounts (BR-001)	"This email already belongs to an account. Use a different email."
Role	Select: Administrator, Manager, Counter Staff	Yes	Disabled after creation (BR-003)	Helper text: "The role cannot be changed later."

Field	Type	Required	Validation	Message shown to user
Assigned branches	Multi-select	Yes for Manager and Counter Staff	At least one active branch	"Assign at least one branch so this person can use the till and boards."
Board color	Palette of 16	Yes for Manager and Counter Staff	Not used by another active employee (BR-007)	"Teal is already used by Amir H. Choose another color."
Initials	Text, 2-3	Yes	Letters; unique among active employees of the assigned branches	"AH is already used at Market Hall. Add a letter, for example AHA."
Temporary password	Secret	Yes on create	BR-006	"Use at least 12 characters. A phrase of three or four words works well."

3.2.2 BRN: Branches & Pickup Slots

Branch configuration, opening and rush hours, order timings, the POS connection, and the slot engine that decides which pickup minutes the kitchen can meet.

Epic: EP-02 Branches & Pickup Slots. **Actors:** Administrator (ADM), Manager (MGR), Customer (CUS), System (SYS). **User stories:** US-008 to US-013.

ID	Requirement	Priority	Release	Roles	Related business rules
FR-BRN-01	The Administrator shall be able to create, edit, reorder and deactivate branches with name, short code, address, tagline and a default-branch flag; a deactivated branch disappears from the customer apps and the till at once.	Must	R1	ADM	—
FR-BRN-02	The Administrator, and a Manager for assigned branches, shall be able to maintain each branch's weekly opening hours with an optional rush window per day.	Must	R1	ADM, MGR	BR-009
FR-BRN-03	The Administrator, and a Manager for assigned branches, shall be able to record closure dates on which no pickup times are offered to customers.	Should	R1	ADM, MGR	BR-009
FR-BRN-04	The Administrator shall be able to set a branch's order timings (online lead time, cancellation cutoff, minutes per kitchen unit, rush factor and online order limit) within the ranges in section 2.7; a Manager can change only the lead time and cancellation cutoff of assigned branches.	Must	R1	ADM, MGR	BR-010, BR-014
FR-BRN-05	The Administrator shall be able to connect each branch to its account at the cloud POS and invoicing provider, see the connection status, and import the branch's cash and card payment methods.	Must	R1	ADM	—
FR-BRN-06	The system shall compute, for a branch, a cart and a channel, the pickup minutes of the current day and mark each one available or unavailable.	Must	R1	CUS, SYS	BR-010, BR-012, BR-020
FR-BRN-07	The system shall compute an order's kitchen minutes from its units in kitchen-time categories, the branch's minutes per unit and the rush factor.	Must	R1	SYS	BR-011
FR-BRN-08	The system shall reserve an order's kitchen minutes atomically when the order is placed, so that two orders can never hold the same kitchen minute at a branch.	Must	R1	SYS	BR-012, BR-013

ID	Requirement	Priority	Release	Roles	Related business rules
FR-BRN-09	A Manager or Administrator shall be able to set and remove one busy window for today at a branch; minutes inside it are unavailable to new orders from the moment it is saved.	Must	R1	MGR, ADM	BR-015
FR-BRN-10	The system shall apply kitchen delay protection while orders are late and show the blocked minutes on the boards, where a Manager can release them early.	Must	R1	SYS, MGR	BR-016
FR-BRN-11	The system shall release an order's unused kitchen minutes when it is canceled, collected or becomes a no-show, and push the changed availability to open pickup-time pickers within 2 seconds.	Must	R1	SYS	BR-017
FR-BRN-12	The system shall cap the kitchen minutes that online orders can hold in each 15-minute rush block, and release the reserved share to online orders before the block starts.	Must	R1.2	SYS, ADM	BR-018

Notes and edge cases:

- **Why minutes, not 15-minute slots.** Discovery showed that orders take 1 to 8 minutes of line time and that customers time their pickup to the minute of their lunch break. Fifteen-minute slots wasted up to 40% of line capacity in the prototype test (DEC-02).
- **One rush window per day.** Saturday dinner has a different window from weekday lunch, so the window is set per weekday. Two windows on one day (lunch and dinner) is TBD-02.
- **Kitchen block is contiguous.** An order's kitchen minutes are the P minutes immediately before its pickup minute (BR-012). A single free minute between two orders can only be used by an order that needs one minute. Appendix B, WE-2, shows this.
- **Rush factor is decided by the pickup minute,** not by the time of ordering. The rush window end is exclusive: with rush 11:45-13:30, a pickup at 13:29 uses the rush factor and 13:30 does not.
- **Busy window validation.** The previous system accepted an end before the start and windows outside opening hours. The platform validates both (field spec below) and keeps the window for today only.
- **Delay protection releases itself.** In the previous system staff had to free blocked minutes by hand and often forgot, so the afternoon looked fully booked. The platform releases blocks automatically when no order is late (BR-016); the manual release stays for Managers.
- **POS connection.** Only cash-type methods and the provider's card method are imported; other methods would not settle through the platform and are ignored with a note on the connection screen.

Field specification: branch order timings and hours (Admin Panel)

Field	Type	Required	Validation	Message shown to user
Open (per weekday)	Checkbox	Yes	Unchecked disables the four times for that day	—
Open time; close time	Time (HH:MM)	Yes when open	open < close	"Closing time must be after opening time (11:00)."
Rush start; rush end	Time (HH:MM)	No	Both or neither; open <= rush start < rush end <= close (BR-009)	"Set both rush times or neither." / "Rush hours must sit inside opening hours (11:00-21:00) and end after they start."
Online lead time (min)	Integer stepper	Yes	5-60	"Enter a lead time between 5 and 60 minutes."

Field	Type	Required	Validation	Message shown to user
Cancellation cutoff (min)	Integer stepper	Yes	0-60	"Enter a cutoff between 0 and 60 minutes before pickup."
Minutes per kitchen unit	Decimal stepper (0.5)	Yes (ADM only)	0.5-5.0	"Enter a value between 0.5 and 5.0 in steps of 0.5."
Rush factor	Decimal	Yes (ADM only)	0.25-1.0	"Enter a rush factor between 0.25 and 1.0."
Online order limit (kitchen units)	Integer	Yes (ADM only)	10-60	"Enter a limit between 10 and 60 units."

Field specification: busy window (boards)

Field	Type	Required	Validation	Message shown to user
Start	Time (HH:MM), today	Yes	Not in the past; within today's opening hours	"Start must be between now (14:05) and closing time (21:00)."
End	Time (HH:MM), today	Yes	After start; at or before closing time	"End must be after the start." / "End must be no later than closing time (21:00)."
Reason	Select: Equipment, Staffing, Supplies, Other	Yes	"Other" needs a note of 5-140 characters	"Add a short note when the reason is Other."
Save	Action	n/a	Replaces any existing window for today (BR-015)	Confirmation: "Replace today's busy window 15:00-15:30 with 16:00-16:45?"

3.2.3 MNU: Menu & Catalog

Categories, products, shared ingredients and add-ons, deposits, POS article mapping, allergen information and daily sold-out status.

Epic: EP-03 Menu & Catalog. **Actors:** Administrator (ADM), Manager (MGR), Customer (CUS), System (SYS). **User stories:** US-014 to US-018.

ID	Requirement	Priority	Release	Roles	Related business rules
FR-MNU-01	The Administrator shall be able to maintain each branch's menu categories: name, image, tile color, display order, active flag and whether the category counts toward kitchen time.	Must	R1	ADM	BR-011
FR-MNU-02	The Administrator shall be able to maintain products per branch: name, description, gross price, category, takeaway and dine-in VAT rates, image, tile color, display order, "show in customer apps" flag, active flag, removable ingredients, available add-ons and an optional deposit product.	Must	R1	ADM	BR-024
FR-MNU-03	The Administrator shall be able to maintain a shared list of ingredients with name, allergens and active flag, and attach ingredients to products as removable; ingredients carry no price.	Must	R1	ADM	BR-021
FR-MNU-04	The Administrator shall be able to maintain a shared list of add-ons with name, gross price, allergens and active flag, and attach add-ons to products.	Must	R1	ADM	BR-022
FR-MNU-05	The system shall support deposit products and show deposits as combined, non-editable lines in carts, orders and invoices.	Should	R1	ADM, SYS	BR-026

ID	Requirement	Priority	Release	Roles	Related business rules
FR-MNU-06	The system shall map every product and add-on to an article in the branch's POS account, create the article when no mapping exists, and push changes to name, price, VAT rate and active status to the POS within 5 minutes.	Must	R1	SYS	—
FR-MNU-07	The system shall hold the 14 EU allergen groups for products, ingredients and add-ons and show the combined allergens of an item before it can be added to the cart in both customer apps.	Must	R1	CUS, ADM	BR-028
FR-MNU-08	A Manager shall be able to mark a product sold out at an assigned branch for the rest of the day; it shows as unavailable in every channel and resets at the start of the next day.	Should	R1	MGR	—

Notes and edge cases:

- **Ingredients have no price (BR-021).** In the previous system an ingredient's price was subtracted when a customer removed it. The owners decided in discovery that removing onions does not reduce labor or cost, and that removal credits did not match the POS articles (DEC-03). Removing an ingredient is free and never reduces the price.
- **Shared ingredients and add-ons, per-branch products.** Ingredients and add-ons are shared by all branches so allergen data is kept once. Products, prices and categories are per branch because the Campus branch sells a smaller menu at student prices.
- **Hidden but sellable.** A product with "show in customer apps" off still appears on the till (for example a staff meal or a counter-only side).
- **Article sync.** The previous system did not push a product's active status to the POS. FR-MNU-06 includes active status, so a retired product cannot be rung up on the POS by mistake.
- **Allergens and removals (BR-028).** Removing an ingredient does not remove its allergens from the display. The sheet says "Removing an ingredient does not make this item free from its allergens", because the line shares equipment.

3.2.4 ORD: Customer Ordering

Branch selection, menu browsing, customization, cart and pricing, pickup time, placement, order history, cancellation and reorder in the two customer apps.

Epic: EP-04 Customer Ordering & Notifications. **Actors:** Customer (CUS), Guest (GST), System (SYS). **User stories:** US-019 to US-025.

ID	Requirement	Priority	Release	Roles	Related business rules
FR-ORD-01	The system shall list active branches in the Administrator's display order with today's opening hours and an open or closed indicator, and remember the customer's selected branch on the device.	Must	R1	CUS, GST	—
FR-ORD-02	The system shall show the selected branch's menu by category with product name, description, price, allergens and availability, showing only active products flagged for the customer apps.	Must	R1	CUS, GST	—
FR-ORD-03	The system shall let a customer customize an item by removing included ingredients and selecting add-ons, set a quantity from 1 to 20, and see the running line total before adding it to the cart.	Must	R1	CUS	BR-014, BR-021, BR-022, BR-023, BR-028

ID	Requirement	Priority	Release	Roles	Related business rules
FR-ORD-04	The system shall keep one cart per customer per branch, let the customer edit lines, change quantities and remove lines, and merge lines whose product and choices are identical.	Must	R1	CUS	BR-026
FR-ORD-05	The system shall price the cart and the order on the server and show net, VAT per rate and total.	Must	R1	CUS, SYS	BR-023, BR-024, BR-025, BR-027
FR-ORD-06	The system shall show the customer the pickup minutes for the cart and explain why none are offered: branch closed, fully booked today, or order too large.	Must	R1	CUS	BR-010, BR-011
FR-ORD-07	The system shall place an order only after re-pricing the cart, confirming that the branch is open, reserving the kitchen minutes and applying the customer's payment choice (pay online now or pay at the counter); Prepay-only customers must pay online.	Must	R1	CUS, SYS	BR-013, BR-027, BR-042
FR-ORD-08	On placement the system shall show a confirmation with the order number, branch, pickup time and cancellation deadline, and send a confirmation email with the same details.	Must	R1	CUS	BR-033
FR-ORD-09	The system shall show the customer's orders of the last 90 days with a date filter, and update each order's status live while the screen is open.	Must	R1	CUS	BR-029
FR-ORD-10	The system shall show an order's detail: items with removed ingredients and add-ons, deposit line, net, VAT per rate, total, tip, payment status and, when paid, the invoice.	Must	R1	CUS	BR-034
FR-ORD-11	The system shall let a customer cancel an order within the cancellation window and show the deadline on the order until it passes.	Must	R1	CUS	BR-017, BR-030
FR-ORD-12	The system shall let a customer reorder a past order at the same branch by copying its lines into the cart at current prices and flagging items that are no longer available.	Could	R1	CUS	—

Notes and edge cases:

- **Prices on the client are display only (BR-027).** The app shows a running total for speed, but the server re-prices at every cart change and at placement. If a price changed in between, placement is refused once with the new total.
- **Pickup minute re-check (BR-013).** Two customers can see the same free minute. The first to place wins; the second sees "12:31 was just taken. Choose another time." with a refreshed list.
- **Order too large (BR-014).** An online cart above 30 kitchen units gets no pickup times and the message "Orders over 30 bowls and wraps need a call to the branch so the kitchen can plan." with the branch phone number.
- **One cancellation boundary.** In the previous system the mobile app used "more than 15 minutes" and the web panel "15 minutes or more". BR-030 sets one rule for both: canceling at exactly 15:00 minutes before pickup is allowed (DEF-022, SRS v1.2).
- **Customer apps show the same history.** The previous web panel showed only today's orders while the app showed 30 days. Both now show 90 days (FR-ORD-09).
- **Order number (BR-033).** The previous reference was derived from the pickup time and was not unique. The platform issues a branch-coded number such as MH-0142 that staff can call out.

Field specification: item customization sheet (both apps)

Field	Type	Required	Validation	Message shown to user
Included ingredients	Checkbox per ingredient, all checked	n/a	Unchecking removes the ingredient; price unchanged (BR-021)	Helper text: "Removing an ingredient does not change the price."
Add-ons	Checkbox per add-on with price	No	Each add-on once per item (BR-022); inactive add-ons hidden	—
Allergens	Read-only chips	n/a	Product, included ingredients and selected add-ons (BR-028)	"Contains: gluten, milk, sesame. Removing an ingredient does not make this item free from its allergens."
Quantity	Stepper	Yes	1-20 (BR-014)	"You can add up to 20 of one item. For larger orders, call the branch."
Add to cart (running total)	Action	n/a	Total = (price + add-ons) x quantity (BR-023); guest must sign in (FR-IAM-03)	Button reads "Add 1 for EUR 11.40".

Field specification: pickup time and placement (both apps)

Field	Type	Required	Validation	Message shown to user
Pickup minute	Grid of minutes, available or unavailable	Yes	Available per BR-010 to BR-012 at the time of display; re-checked at placement (BR-013)	"Choose a pickup time." / "12:31 was just taken. Choose another time."
No times offered	Empty state	n/a	Reason from the slot engine	"Market Hall is closed today." / "Today's pickup times at Market Hall are fully booked." / "Orders over 30 bowls and wraps need a call to the branch."
Payment choice	Radio: Pay online now; Pay at the counter	Yes	Prepay-only accounts see only Pay online now (BR-042)	"Your account needs online payment until 2026-10-16 because a ready order was not collected."
Place order	Action	n/a	Re-price (BR-027), branch open, kitchen minutes reserved (BR-013)	"The total changed to EUR 34.50 because a price was updated. Check your cart and place the order again."

3.2.5 PAY: Payments & Invoicing

Online payment with tip, counter settlement by cash or card, single-capture payment attempts, invoices and cancellation receipts in the POS account, refunds and nightly reconciliation.

Epic: EP-05 Payments & Invoicing. **Actors:** Customer (CUS), Counter Staff (CST), Manager (MGR), Administrator (ADM), System (SYS). **User stories:** US-030 to US-035.

ID	Requirement	Priority	Release	Roles	Related business rules
FR-PAY-01	The system shall take online payments through the card payment provider's hosted checkout, including strong customer authentication, for the order total plus the selected tip.	Must	R1	CUS	BR-038, BR-042
FR-PAY-02	The system shall offer online tip options of 0, 5, 10 and 15% with 0% preselected, and show the amount to pay before checkout.	Should	R1	CUS	BR-036

ID	Requirement	Priority	Release	Roles	Related business rules
FR-PAY-03	The system shall mark an online payment Succeeded only from the provider's signed webhook, process each provider event once, and update the order on every channel within 2 seconds of the webhook.	Must	R1	SYS	BR-035, BR-038
FR-PAY-04	The till shall settle an unpaid order by cash (press-and-hold control) or by card through the card reader, with an optional tip keyed as the total amount on the card path.	Must	R1	CST, MGR	BR-035, BR-036
FR-PAY-05	The system shall manage payments as attempts with an idempotency key so that an order is charged at most once, and a retried confirmation never starts a new charge.	Must	R1.1	SYS	BR-035
FR-PAY-06	After a payment succeeds, the system shall raise one invoice in the branch's POS account, retrying through an outbox until the POS confirms, and store the invoice number and PDF link.	Must	R1	SYS	BR-025, BR-037
FR-PAY-07	The system shall let a customer view and download the invoice PDF of a paid order in both customer apps.	Must	R1	CUS	BR-037
FR-PAY-08	The system shall refund a canceled online-paid order in full through the provider's refund API, and route refunds that still fail after 3 retries to a Manager queue.	Must	R1	SYS, MGR	BR-030, BR-031
FR-PAY-09	The system shall raise a cancellation receipt in the POS account when an invoiced order is canceled, and let a Manager record a counter refund with method and reference.	Must	R1	SYS, MGR	BR-031
FR-PAY-10	The system shall reconcile the previous day's provider transactions, the platform payments and POS invoices each night and show mismatches to Administrators.	Must	R1.1	SYS, ADM	BR-039

Notes and edge cases:

- **Pay online at any time before collection.** An order placed "pay at the counter" can still be paid online from the order card until it is collected or canceled. The previous web panel had no online payment; both apps now do.
- **Counter staff cannot refund.** Counter Staff can cancel unpaid orders only. Canceling a paid order needs a Manager because it moves money (BR-031, DEC-06).
- **Card reader timeouts (CR-007).** When the reader or the confirmation call times out, the till never offers "pay again". It offers "Verify payment", which asks the provider about the same attempt (BR-035). This closed the cause of INC-2026-009.
- **Online payment pending at the counter.** If a customer started an online checkout and then wants to pay cash, the till shows "Online payment in progress". A Manager can expire the checkout through the provider and then settle at the counter.
- **Tips (BR-036).** Tips are recorded apart from the order total, appear as a separate line on the invoice, and are outside the VAT calculation (A-04).
- **Invoices are asynchronous (CR-007).** A paid order shows "Invoice being prepared" until the POS confirms, normally within 10 seconds. An outage at the POS provider never blocks payment (NFR-REL-04).

Field specification: counter card payment with tip (till)

Field	Type	Required	Validation	Message shown to user
Amount to charge	Numeric keypad (EUR)	Yes	Valid amount; at least the order total; at most 1.5 x the total (BR-036)	"Enter an amount, for example 20.00." / "The amount must be at least the order total, EUR 18.60." / "That tip is more than half the order. Check the amount."
No tip	Action	n/a	Charges the order total	—
Card reader status	Read-only	n/a	Reader paired and merchant signed in	"Card reader not connected. Turn it on and tap Retry, or take cash."
Verify payment	Action (after a timeout)	n/a	Queries the same attempt; never starts a new charge (BR-035)	"Checking the card payment with the provider. Do not charge the card again."

3.2.6 TIL: Till & Live Order Boards

The iPad till, the live queue, the kitchen and counter boards in the Admin Panel, board display rules, live updates and resync, staff alerts and the daily operations summary.

Epic: EP-06 Till & Live Order Boards. **Actors:** Counter Staff (CST), Manager (MGR), Administrator (ADM), System (SYS). **User stories:** US-036 to US-042.

ID	Requirement	Priority	Release	Roles	Related business rules
FR-TIL-01	The till shall run in landscape on iPad with three panels: product tiles grouped by category in the menu's colors, the order being built, and the live queue of the branch.	Must	R1	CST, MGR	—
FR-TIL-02	The till shall let staff add products through a customization dialog (ingredients, add-ons, quantity 1 to 20) and choose Take away or Dine in; dine-in orders exist only on the till.	Must	R1	CST, MGR	BR-014, BR-032
FR-TIL-03	The till shall preselect the earliest free pickup minute once the first item is added, let staff choose another, and offer overbooked minutes when none are free.	Must	R1	CST, MGR	BR-019
FR-TIL-04	The till shall place an order unpaid from the time field, or place and settle it in one action from a payment button, and keep the order being built per device.	Must	R1	CST, MGR	—
FR-TIL-05	The till shall show the branch's live queue from all channels, let staff call up an unpaid order to change or settle it, and mark a paid order Collected with Deliver.	Must	R1	CST, MGR	BR-029, BR-034
FR-TIL-06	The till shall let staff cancel an order with a press and hold and a reason.	Must	R1	CST, MGR	BR-031
FR-TIL-07	The Admin Panel kitchen board shall show the branch's open orders by pickup time with items, removed ingredients, add-ons, order type and status, without customer contact data, and let staff mark an order Ready.	Must	R1	MGR, ADM	BR-029, BR-032
FR-TIL-08	The Admin Panel counter board shall show the same orders with the customer's name and contact details, payment status and method, and let staff mark an order Ready or cancel it with a reason.	Must	R1	MGR, ADM	BR-031

ID	Requirement	Priority	Release	Roles	Related business rules
FR-TIL-09	Boards and the till queue shall show who took an order and its state: a tint for dine-in or the employee's color with initials, and a text status badge.	Must	R1	CST, MGR, ADM	BR-007, BR-020
FR-TIL-10	The system shall propagate every order change to the boards, tills and the customer's app within 2 seconds at the 95th percentile, scoped to the branch.	Must	R1	SYS	BR-020, BR-045
FR-TIL-11	Boards and tills shall show a live-connection indicator and, after a reconnect or a sequence gap, re-subscribe and load a branch snapshot before showing further changes.	Must	R1.1	SYS	BR-045
FR-TIL-12	The boards shall play a sound and show a message for new, changed and customer-canceled orders, refresh silently for status changes, and keep the last 50 alerts of the session in an alert list.	Should	R1	MGR, ADM	—
FR-TIL-13	The Admin Panel shall show a daily operations summary per branch and channel (orders, ready on time, late, canceled, no-shows, overbooked, sales) with CSV export.	Should	R1	ADM, MGR	—

Notes and edge cases:

- **Drafts per device.** In the previous system two tills signed in with the same account shared one draft and overwrote each other. The platform keeps the draft per device and per account (DEF-038).
- **Called-up orders are locked.** An order called up on one till disappears from other tills' queues and shows "Open on Till 2 (AH)" on the boards until it is saved or released (BR-034).
- **Till cancellations are not deletions.** The previous till removed canceled orders permanently and left their invoices outstanding. The platform keeps every canceled order with reason and actor, and raises a cancellation receipt where needed (BR-031).
- **Two-channel display.** The previous boards used one color per row with a precedence order, so a late till order showed the employee's color and its lateness was hidden. The platform separates the tint (who or what) from the status badge (Queued, Preparing, Late, Ready, Canceled) (business rules table 3.4).
- **Live indicator and resync (CR-004).** INC-2026-004 showed that a till could reconnect after a network drop without re-joining its branch room and then show a frozen queue for 40 minutes with no warning. FR-TIL-11 makes staleness visible and self-healing.
- **Alert sound.** Browsers block sound until the user interacts with the page. Board mode starts with a "Start board" button, which also unlocks sound.

Field specification: staff cancellation (till and counter board)

Field	Type	Required	Validation	Message shown to user
Reason	Select: Customer request, Item unavailable, Duplicate order, Kitchen problem, Other	Yes	"Other" needs a note of 5-140 characters	"Choose why this order is canceled."
Refund summary	Read-only	n/a	Shows refund amount and route (online automatic, counter manual)	"EUR 22.68 will be refunded to the card ending 4421."
Permission	n/a	n/a	Counter Staff cannot cancel paid orders (BR-031)	"Only a Manager can cancel a paid order. Ask your Manager."

Field	Type	Required	Validation	Message shown to user
Confirm (press and hold 0.6 s)	Action	n/a	Order not Collected	"Order MH-0142 was already collected and cannot be canceled."

3.2.7 NTF: Notifications & Uncollected Orders

Push registration, status notifications, the uncollected-order reminder ladder, the no-show cutoff and transactional email.

Epic: EP-04 Customer Ordering & Notifications. **Actors:** Customer (CUS), System (SYS). **User stories:** US-026 to US-029.

ID	Requirement	Priority	Release	Roles	Related business rules
FR-NTF-01	The system shall register a push token per device for signed-in customers in both apps after the customer allows notifications, and remove only that device's token on sign-out.	Must	R1	CUS	BR-043
FR-NTF-02	The system shall send a push notification when an order is marked Ready, canceled by staff, or refunded.	Must	R1	CUS	—
FR-NTF-03	The system shall run the uncollected-order reminder ladder for Ready, unpaid pickup orders.	Must	R1	SYS	BR-040
FR-NTF-04	The system shall mark uncollected Ready orders as No-show at the cutoff and apply the Prepay-only status to the account, emailing the customer.	Must	R1	SYS	BR-041
FR-NTF-05	The system shall send transactional email (sign-in codes, order confirmations, reminders, no-show notices, staff password reset) through the transactional email provider in the customer's chosen language.	Must	R1	SYS	BR-044
FR-NTF-06	The system shall log every notification with event, recipient, channel, status and attempts, and deduplicate and retry sends.	Must	R1	SYS	BR-044
FR-NTF-07	Tapping a notification shall open the related order in the app and refresh the order list.	Should	R1	CUS	—

Notes and edge cases:

- **Anchor for the ladder (BR-040).** The ladder counts from the later of the pickup minute and the ready time, so a customer is never chased because the kitchen was late.
- **No-show consequence changed in R1.1 (CR-003).** In R1 an unpaid no-show suspended the account, and only an Administrator could reinstate it. In the first three weeks, 23 accounts were suspended and 9 suspensions were disputed, mostly because staff had handed the order over without marking it. Since R1.1, a no-show sets Prepay-only for 30 days instead.
- **Order confirmation email.** The previous system sent nothing at placement. The confirmation email (FR-ORD-08) states the cancellation deadline, which discovery linked to fewer disputes.
- **Content.** Push and email carry the order number, branch and pickup time only, never items or payment data (NFR-PRIV-01).

3.3 Role-permission matrix

Permissions are enforced server-side (FR-IAM-06, BR-003). "Assigned" means the Manager's or Counter Staff member's assigned branches.

Capability	ADM	MGR	CST	CUS	GST
Browse branches and menus	Yes	Yes	Yes (till)	Yes	Yes
Place a customer order (app, web)	No	No	No	Yes	No (sign in)
Cancel own order within the window	n/a	n/a	n/a	Yes	n/a
Take till orders, dine-in, settle by cash or card	No (refused at till sign-in, BR-004)	Assigned	Assigned	No	No
Cancel unpaid order (till, counter board)	Yes (board)	Assigned	Assigned (till)	No	No
Cancel paid order and record counter refund	Yes (board)	Assigned	No	No	No
Mark Ready (kitchen and counter boards)	Yes	Assigned	No	No	No
View customer contact details (counter board)	Yes	Assigned	No	Own only	No
Set busy window, release delay block, mark sold out	Yes	Assigned	No	No	No
Edit opening hours, rush hours, lead time, cutoff	Yes	Assigned	No	No	No
Edit kitchen factors, online limit, rush throttle	Yes	No	No	No	No
Create or deactivate branches; POS connection	Yes	No	No	No	No
Maintain categories, products, ingredients, add-ons	Yes	No	No	No	No
Manage employees	Yes	No	No	No	No
Manage customer accounts; clear Prepay-only	Yes	No	No	No	No
Daily operations summary	All branches	Assigned	No	No	No
Reconciliation report; online payment settings	Yes	No	No	No	No

3.4 Quality requirements

The 28 NFRs are in non-functional-requirements.md, classified by ISO/IEC 25010. The ones that most shape the design:

NFR	Summary	Why it matters here
NFR-PERF-01	Pickup-slot list in 800 ms or less at p95	Customers choose a time in a lunch break
NFR-PERF-03	Order change on every screen within 2 s at p95	The kitchen works from the board, not from paper

NFR	Summary	Why it matters here
NFR-REL-02	Current state within 5 s of reconnect; stale state never shown without a warning	INC-2026-004
NFR-REL-04	POS outage never blocks ordering or payment	Fiscal invoices are raised outside the platform
NFR-SEC-03	No card data in the platform systems	Keeps PCI DSS scope to SAQ A for e-commerce
NFR-USE-02	Walk-in till transaction 50 s median or less	OBJ-06
NFR-ACC-02	Board status never by color alone	Kitchen legibility and WCAG 1.4.1
NFR-OBS-02	Business anomaly alerts, including double capture	INC-2026-009

3.5 Data requirements

Entities, fields and retention are in the data dictionary and the ERD. Key rules:

- Money is stored as integer cents with the currency EUR. VAT rates are stored as integer per-mille (100 = 10.0%).
- An order stores a priced snapshot of every line (product name, unit price, add-ons, removed ingredients, VAT rate). Later catalog changes never change a placed order.
- Orders, payments, invoices and refunds are never deleted. Customer personal data is erased or anonymized per BR-008 and NFR-PRIV-02.
- Kitchen reservations are stored one document per kitchen minute with a unique index on branch, date and minute (ADR-001).

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4. Verification

Requirement class	Primary method	Evidence
Functional requirements	Test: system test cases traced in the RTM	test-cases.csv, CI reports
Business rules	Test: automated unit or API test per rule (NFR-MNT-01); decision-table rows as test data	CI coverage gate
Worked examples WE-1 and WE-2	Test: canonical cases TC-ORD-005 and TC-BRN-003, run on every release	Featured cases in test-cases.md
Performance and reliability NFRs	Test: load test at the peak profile (NFR-PERF-05); chaos test for socket drops	Load and chaos reports per release
Security and privacy NFRs	Inspection and test: OWASP ASVS L2 checklist, external penetration test, DPIA review	Pen test report, DPIA
Accessibility NFRs	Inspection and test: automated checks plus manual screen-reader passes	Accessibility report
Usability NFRs	Demonstration: timed tasks in UAT (UAT plan)	UAT sign-off

Appendix A. Open issues

Each row records the behavior that R1.2 implements today, so nothing ambiguous is left implicit.

ID	Issue	Behavior in R1.2	Owner	Target
TBD-01	Board-only role for the kitchen display	The kitchen display runs in board mode under the shift Manager's session, with a 14 h maximum	Product Owner	R2
TBD-02	Two rush windows on one day (lunch and dinner)	One rush window per weekday	Product Owner	R2
TBD-03	Partial refunds (one item of an order)	Full refunds only; partial adjustments are done in the POS by a Manager	Finance Controller	R2
TBD-04	Crash reporting in the customer apps	App store consoles only; error monitoring on the till	Tech Lead	R2
TBD-05	Tax treatment of tips if national rules change	Tip as a separate line outside VAT (A-04, confirmed 2026-05-14)	Finance Controller	Review yearly
TBD-06	Order-ahead for another day	Today only	Product Owner	Not planned
TBD-07	Separate online merchant account per branch	One merchant account for the single legal entity	Finance Controller	On any change of ownership
TBD-08	Second production line at Market Hall	One line; the rush factor stands in for extra staff	Operations Director	Q1 2027 review
TBD-09	Reorder when an add-on is retired	The add-on is dropped and the line is flagged "changed"	Product Owner	R2
TBD-10	Till offline mode during an internet outage	No offline mode; runbook switches the counter to the POS provider's own app	Tech Lead	R2

Appendix B. Worked examples

Both examples use the same canonical order, MH-0142, placed by PER-01 Clara Mendes at LOC-01 Market Hall on Friday 2026-09-18. The numbers here are identical in business-rules.md, US-021, US-011 and the test cases.

WE-1. Order total with add-ons, removed ingredients, VAT and rounding

Line	Item	Base	Add-ons	Removed (no credit)	Unit price	Qty	Line total	VAT rate
1	Chicken Grill Bowl	9.40	Halloumi 1.50; Garlic sauce 0.50	Red onion; Coriander	11.40	1	11.40	10%
2	Chicken Wrap	7.90	Extra chicken 2.20	Pickled chili	10.10	2	20.20	10%
3	Sparkling lemonade 0.33 l	2.90	—	—	2.90	1	2.90	20%

VAT rate	Gross	VAT = gross x r / (100 + r), half up	Net
10%	31.60	31.60 × 10 / 110 = 2.8727 → 2.87	28.73

VAT rate	Gross	VAT = gross x r / (100 + r), half up	Net
20%	2.90	2.90 × 20 / 120 = 0.4833 → 0.48	2.42
Order	**34.50**	**3.35**	**31.15**

- Removing Red onion, Coriander and Pickled chili changes nothing (BR-021).
- Per-line VAT would give 1.04 + 1.84 + 0.48 = 3.36, one cent more than the POS invoice. BR-025 extracts VAT per rate group, which matches the invoice.
- Clara pays online with a 5% tip: 34.50 × 5 / 100 = 1.725, rounded half up to **1.73** (BR-036). Half-even rounding would give 1.72. The card is charged **EUR 36.23**. The invoice shows the tip as a separate line outside VAT.

WE-2. Pickup-slot calculation during the rush hour

Settings at Market Hall: rush 11:45-13:30, online lead time L = 10 min, m = 1.0 minute per kitchen unit, rush factor f = 0.5. Clara opens the pickup-time sheet at 12:04:20.

- Kitchen units U = 3 (one bowl and two wraps; drinks do not count toward kitchen time).
- For any pickup minute from 11:45 to 13:29, P = ceil(3 × 1.0 x 0.5) = ceil(1.5) = **2** kitchen minutes. For 13:30 or later, P = 3.
- Earliest pickup = 12:04:20 + 10 min = 12:14:20, rounded up to **12:15**.

Kitchen minutes already held:

Order	Channel	Pickup	Units	P (rush)	Kitchen minutes held
MH-0131	App	12:15	6	3	12:12-12:14
MH-0133	App	12:18	5	3	12:15-12:17
MH-0134	Till	12:19	2	1	12:18
MH-0136	Web	12:23	7	4	12:19-12:22
MH-0138	App	12:26	3	2	12:24-12:25

Pickup T	Kitchen minutes needed (T - 2 to T - 1)	Result
12:15 to 12:23	12:13-12:14 up to 12:21-12:22	Unavailable: every pair overlaps MH-0131, MH-0133, MH-0134 or MH-0136
12:24	12:22-12:23	Unavailable: 12:22 held by MH-0136
12:25	12:23-12:24	Unavailable: 12:24 held by MH-0138. The single free minute 12:23 cannot hold a 2-minute order
12:26, 12:27	12:24-12:25, 12:25-12:26	Unavailable: held by MH-0138
12:28	12:26-12:27	**First available minute**

Clara chooses **12:31**. Order MH-0142 holds kitchen minutes 12:29-12:30, and its cancellation deadline is **12:16** (12:31 minus 15 minutes). The R1.2 rush throttle (BR-018) does not change this result: the 12:15 block's reserve was released at 11:55, and in the 12:30 block MH-0142 is the first online order (1 of 9 online minutes).

Related documents

- Business Requirements Document
- Business rules
- Non-functional requirements
- Compliance mapping
- Glossary
- Requirements traceability matrix
- System architecture
- Data dictionary
- OpenAPI contract
- Epics
- Change request log
- Test cases